



Research Paper

Numerical investigation of tip clearance designs for oblique-cut fin microchannels

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ARTICLE INFO

Keywords:

Enhanced microchannel
Oblique-cut fin
Tip clearance
Heat transfer
Electronic cooling

ABSTRACT

Tip clearance represents a simple design intervention strategy to mitigate the high hydraulic penalties often associated with microchannel heat sinks. This approach has not been widely explored for enhanced microchannel heat sinks, having previously focused only on conventional (uniform) tip clearance design for straight microchannels. In view of this, the present work investigates the performance of an enhanced heat sink, specifically oblique-cut fin microchannels, with tip clearance designs. Two different designs are considered, namely conventional/uniform tip clearance (*utc*) and a newly conceptualized modulated tip clearance (*mtc*). Numerical simulations are conducted using ANSYS FLUENT 2022 R1 first for straight microchannels, which are considered the baseline to which oblique-cut fin microchannel designs are compared. The simulations are carried out under laminar flow regime including conjugate heat transfer in a heated solid substrate for flow rates of 250 – 650 mL/min. The results indicate that *utc* significantly reduces pressure drop though can increase thermal resistance for the oblique channels. The underlying physical mechanisms responsible for this observation are elucidated by examining thermal and flow behaviours in and around oblique cut slots. Based on the flow behaviour in oblique slots with *utc*, modulated tip clearance (*mtc*) concept is introduced and investigated. When compared to *utc* cases, *mtc* tends to reduce the thermal resistance substantially, though depending upon oblique cut angle and flow rate. However, the Performance Index (*PI*) shows that *utc* improves the overall performance in the range of 10–15 % compared to that observed with *mtc* cases, though the factor of improvement reduces closely to 5 % at 650 mL/min. In addition, the overall performance in terms of heat transfer and hydraulic pumping power further show that *utc* and *mtc* design strategies are competitive, and warrant further multi-objective optimization investigations to determine the best design option overall.

1. Introduction

Recent advancements in semiconductor electronics fabrication have facilitated the miniaturization of electronic components and packages, including but not limited to CPUs, graphics cards and IGBT modules. Simultaneously, the evolution of complex AI and Machine Learning algorithms has spurred the demand for rapid computing/processing capabilities. These progressions have led to a substantial increase in both the power and heat flux generation in electronic components and packages, which if not managed adequately, can lead to performance deterioration and reduced long term reliability. Presently, high performance CPUs can generate heat fluxes above 50 W/cm², with some surpassing 100 W/cm², which are beyond the limits (<40 W/cm²) of

traditional fan-fin air cooling solutions for typical CPU packages [1–3]. To ensure the safe and reliable operation of high heat flux semiconductor electronics, the development of high-performance thermal management systems is imperative since they ultimately determine the operational temperature of these devices. Given the compact nature of these devices and systems, liquid-based microchannel heat sinks, initially proposed by Tukerman and Pease [4], have garnered significant attention over the past several decades. These heat sinks feature micro-scale channels with hydraulic diameters typically at or below 300 μm [5,6] and offering superior heat dissipation solutions for electronics cooling applications compared against air-based cooling technologies. Their effectiveness stems from the large surface area-to-volume ratios and high convective heat transfer coefficients associated with liquid flow in small hydraulic diameters channels, which together lead to

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Received 22 June 2024; Received in revised form 30 August 2024; Accepted 15 September 2024

Available online 18 September 2024

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Nomenclature			
C_p	Specific heat (J/kg K)	<i>Obl</i>	Oblique fin microchannel
D_h	Hydraulic diameter (m)	<i>tc</i>	Tip clearance
I	Current (A)	<i>utc</i>	Uniform tip clearance
k	Thermal conductivity (W/m K)	<i>mtc</i>	Modulated tip clearance
L, l	Length (mm)	<i>Greek symbols</i>	
m'	Mass flow rate (kg/s)	θ	Angle (°)
N	Number of primary channels	ρ	Density (kg/m ³)
p	Pitch (mm)	μ	Kinematic viscosity (m ² /s)
P	Pressure (Pa)	δ	Uncertainty
PI	Performance index	ϕ	Viscous dissipation term (W/m ³)
q'	Heat power (W)	<i>Subscripts</i>	
Re	Reynold's number	<i>b</i>	Base substrate
T	Temperature (K)	<i>c</i>	Channel
t	Thickness and/or height (mm)	<i>exp</i>	Experimental
v	Velocity (m/s)	<i>f</i>	Fin/Channel wall, fluid
V	Voltage (V)	<i>in</i>	Inlet
V'	Volumetric flow rate (m ³ /s)	<i>m</i>	Manifold
W, w	Width (mm)	<i>out</i>	Outlet
<i>Str</i>	Straight microchannel	<i>sim</i>	Simulation

impressively high overall heat transfer rates. However, the pressure drop must be considered carefully in order to manage the hydraulic power penalty and/or to not exceed design operating pressures.

Since the advent of the straight microchannel heat sink concept, researchers have explored various configurations to improve their thermal-hydraulic performance. Investigations have explored geometrical as well as topological features, including but not limited to channel shapes and dimensions [5–7], patterned roughness [8,9], protrusion and mixers [10–12], channel patterning [13,14], and flow configurations [15–19]. Among the numerous enhanced microchannel configurations studied, oblique microchannels have consistently been proven to be one of the higher potential solutions owing to their low thermal resistance at moderate pressure drop, coupled with the straight-forwardness of manufacturing.

Oblique microchannels feature channels/slots oriented at an angle relative to the main flow channels, periodically created along the channel walls. Lee et al., [20] first introduced the concept of oblique slot microchannels and numerically investigated its heat transfer performance. The authors determined that the combination of two phenomena contributed to the observed performance enhancement; periodic breakage and redevelopment of thermal boundary layer along the main channel walls, and enhanced fluid mixing caused by breaching the continuous fins into oblique slots. Impressive overall heat transfer coefficients up to 55 kW/m²K were achieved at a moderate flow rate and pressure drop. Fan et al. [21] performed experimental and numerical investigations on the thermal- hydraulic performance of oblique fins on a cylindrical surface across a range of Reynolds numbers (50–840) and reaffirmed the merits of oblique channels for heat transfer.

Lee et al. [22] conducted a parametric experimental study to explore the impact of oblique fin angle and fin pitch for different Reynolds numbers (180–680) and two hydraulic diameter channels (100 μ m and 200 μ m). Based on the measured test chip temperature and pressure drop across the channel, the authors suggested that smaller oblique angle and smaller fin pitch provide maximum heat transfer enhancement, albeit at the expense of increased pressure drop and pumping power. The pressure drop was comparable to that of straight microchannels up to $Re = 500$, beyond which it increased drastically. They determined that the microchannel with smallest channel diameter (100 μ m), smallest oblique angle (27°), and smallest fin pitch (400 μ m) outperformed other tested combinations. However, it should be noted here that such small dimensions put some practical constraint on micro-

fabrication techniques. Zhang et al. [23] numerically examined the performance of two configurations of oblique microchannels – aligned and staggered. They observed that the staggered configuration performs better at low flow rates (<3.5 g/s), though its merits diminish at high flow rates due to an escalation in the pressure drop and hydraulic pumping power. Through a parametric study, they demonstrated that microchannels with the highest oblique angle tested (60°) outperformed the other microchannels in the staggered configuration.

The published literature has shown that oblique microchannels provide a practical means to enhance the heat transfer over straight microchannels, though can suffer from higher hydraulic penalties. Clearly, striking the balance between the heat transfer enhancement and pressure drop/pumping power demand is critical. This balance can be achieved by reducing the size of geometric features, such as fin width and fin pitch. However, as mentioned earlier, this may then pose some challenges with respect to micro-fabrication processes at lower length scales, along with other practical issues like fouling and clogging, motivating the exploration of alternative engineering solutions that are both fabrication-friendly, long term reliable, and effective in terms of enhancing overall system performance. In this context, the addition of tip clearance has emerged as a straight-forward technical solution, as it is known to influence both heat transfer and pressure drop characteristics of straight microchannels, whilst not adding to the manufacturing cost and complexity, since these are associated with the microchannels themselves [24].

Several studies have underscored the importance of tip clearance in microchannel heat sink performance. Min et al. [25] conducted a numerical investigation into the effect of tip clearance on thermal resistance of straight microchannels at fixed pumping power. Their analysis determined that tip clearance led to 3.5 % enhancement in thermal resistance compared to the optimized microchannel heat sink without tip clearance, and this occurred when the tip clearance was 60 % the channel width. Reyes et al. [24] experimentally found that a moderate decrease in the heat transfer performance was augmented with significant decrease in the hydraulic penalty with carefully engineered tip clearance in straight microchannels. They highlighted the advantage of tip clearance by determining that it can reduce the pumping power by half whilst maintaining the same thermal efficiency as that with the case of no tip clearance. Liang et al. [26] experimentally assessed the effectiveness of tip clearance for straight microchannels using non-Newtonian polymeric fluid. Their findings indicated that, while tip

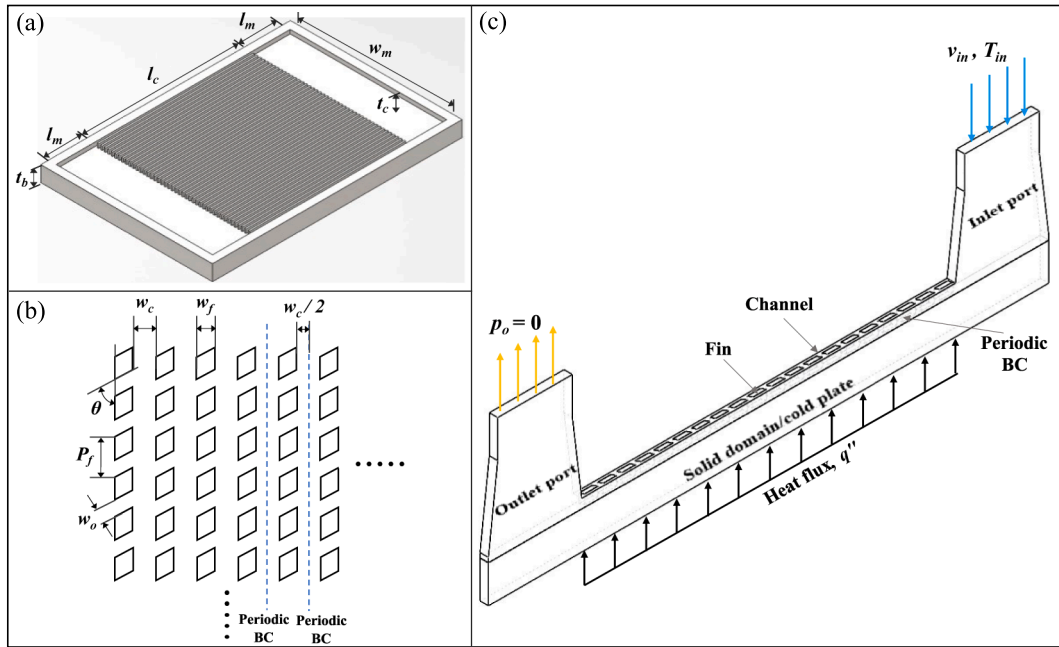


Fig. 1. A) solid model of microchannel heat sink, b) schematic illustrating geometrical details of oblique fin microchannels, and c) computational domain with boundary conditions.

clearance significantly reduced pressure drop, the overall performance was moderated by a minor reduction in overall convective heat transfer intensity, though it could still result in a 20–50 % increase in the figure of merit.

Tabkhi et al. [27] investigated the effect of tip clearance on local heat transfer downstream of a single micro-pin fin in the laminar flow regime. Utilizing conjugate 3D CFD modelling, they showed that the local heat transfer coefficient downstream of the pin fin depended on tip clearance, reaching a maximum at an optimal gap height. The tip clearance was found to influence flow structures thus promoting three dimensionality and mixing, shortening the wake region, and increasing the velocity downstream of the pin fin. Other studies investigating tip clearance for micro-pin fin array type heat sinks showed augmented overall system performance, wherein tip clearance influenced heat transfer and hydrodynamic characteristics differently for different shapes and configuration of pin-fins [24,28,29]. For example, Bhandari and Prajapati [30] performed experiments with a fixed overall channel height and a range of pin fin heights to create tip clearances from 0 % to 50 % of the channel width. Albeit at significantly lower pressure drop, the 50 % case significantly reduced the overall heat transfer coefficient largely due to the loss of heat transfer surface area. However, the 25 % case produces enhanced cooling at lower pressure drop, despite the reduction in overall surface area. This highlights that careful attention must be paid to convective intensity, surface area density, and hydraulic penalties when implementing tip clearance as a potential enhancement strategy.

Considering the above discussion, both oblique fins and tip clearance show high potential as practical design strategies to enhance the thermal hydraulics of microchannel heat sinks. However, combining tip clearance with oblique channels has yet to be explored, which forms the research gap in the open literature, and this must be done so carefully considering the observed sensitivity of the thermal hydraulics to the set of geometric design parameters. Notably, this investigation becomes quite cumbersome, making numerical simulation a suitable initial approach to validate the concept. Subsequently, experimental investigations can be conducted to assess its practical performance. In this context, the present work aims to numerically investigate the impact of conventional/uniform tip clearance design on oblique microchannels, with it also being studied on straight microchannels to provide a

Table 1

Geometrical parameters associated with straight and oblique microchannel heat sinks.

Parameter	Straight microchannels	Oblique microchannels
$l_c \times w_c \times t_c$	20 mm $\times$ 0.250 mm $\times$ 0.500 mm	
$l_m \times w_m$	5 mm $\times$ 20 mm	
t_b	2 mm	
N	45	
w_f	0.200 mm	
w_o	–	0.250 mm
p_f	–	1 mm
θ	–	Variable

baseline for comparison. The fundamental field quantities such as spatial distributions of velocity and temperature along the channels are presented and discussed to examine the influence of tip clearance. Further to this, a new design strategy, here termed modulated tip clearance (*mtc*), is introduced to partially addresses the shortcomings of uniform tip clearance design across the channels. The present work becomes important as it introduces a tip clearance strategy for enhanced heat sinks based on oblique microchannels, aiming to enhance the thermal hydraulic characteristics of these heat sinks. Additionally, it proposes a new tip clearance design with high potential for both straight and oblique channels, paving the way to further explore tip clearance designs for various microchannel types. The findings of the present work highlight that the effectiveness of *utc* and *mtc* varies with the type of microchannel and the geometric parameters. Therefore, the feasibility of tip clearance designs should be further explored with systematic multi-objective optimization. In addition, as observed in the results that the tip clearance significantly reduces pressure drop, further research can be directed to examine the feasibility of tip clearance in microchannels for applications involving highly viscous fluids.

2. Methods

2.1. Computational modelling

A three-dimensional fluid flow and conjugate heat transfer

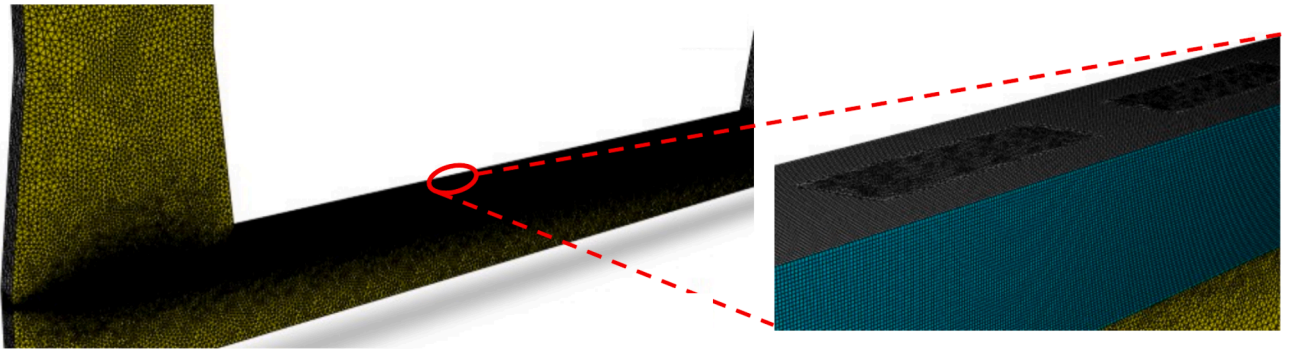


Fig. 2. Grid generation in the computational domain.

simulation model is developed using ANSYS FLUENT 2022 R1 to investigate flow and heat transfer characteristics of straight and oblique fin microchannels, both with and without tip clearance. The geometry of the oblique microchannels considered is shown in Fig. 1 and the associated geometrical parameters are presented in Table 1. It is understood that the straight channel is the same configuration though without the oblique cuts along the channel wall. As can be seen in Fig. 1(a), microchannels are generated on the central portion of the base substrate with small inlet and outlet manifolds. The geometrical parameters of the base plate, manifolds and the channels are defined in Fig. 1(a) and 1(b). The manifolds are connected to the corresponding inlet and outlet ports, oriented normal to the primary flow channels, as shown in Fig. 1(c). Both ports are designed with increasing cross-sectional area towards manifold to minimize the flow mal-distribution.

It is important to mention here that modelling the entire geometry of a microchannel heat sink with multiple parallel channels (Fig. 1(a)) has not been considered here, since the computational domain would require unnecessarily large mesh grid points and associated computational resources and time. Hence, only a section of the geometry is chosen as the computational domain, as shown in Fig. 1(b) and 1(c), exploiting the transverse periodicity of the primary channels and secondary channels (oblique slots). The dotted lines in Fig. 1(b) show transverse boundaries of the computational domain. Thus, simulations are performed with a periodic flow and thermal boundary conditions applied at left and right boundaries. The geometry of the computational domain is created in SolidWorks and the mesh is created using ANSYS meshing tool on Intel(R) Octa Core (TM) i7-10700 CPU @ 2.90 GHz, 64 GB RAM workstation. As shown in Fig. 1(c), the computational domain contains both solid and fluid subdomains. The solid domain is a metallic base substrate volume in which the microchannels are created. The fluid domain encompasses flow regions passing through ports, manifolds, and channels. To accurately capture velocity and temperature gradients adjacent to channel walls, a uniform structured mesh is generated in the channel's fluid domain using a Body Fitted Cartesian method. The Cartesian structured mesh also helps realize the periodic boundary condition employed in the present simulation. The grid generation in the computational domain is shown in Fig. 2. An unstructured tetrahedral mesh with variable mesh sizing is generated using the Patch Conforming method for the base substrate and the ports fluid domain, considering their more complex geometries. It is noted that the mesh size is varied in such a way so as to obtain minimum cell sizes towards the channel domain. Moreover, at various interfaces between the channel domain and the remaining domains, the local mesh size is controlled by the Contact Sizing option to match the cell sizes at these interfaces.

The primary assumptions considered in the present simulations include incompressible, laminar flow, Newtonian fluid, no-slip condition at channel walls, negligible effects of gravity and other unaccounted body forces, and negligible radiative heat transfer. With these assumptions, continuity, momentum, and energy equations defined respectively in Eqs. (1–4) are numerically solved to estimate the steady state velocity,

Table 2

Range of oblique fin angle and tip clearance considered for present simulation.

Configuration	Oblique angle, θ	Tip clearance, t_{tc}	Nomenclature*
Straight microchannels	—	100, 150 and 200 μm	$Str_X_t_{tc}$
Oblique microchannels	40° and 60°		$Obl_ \theta_ X_ t_{tc}$

* X = Variant of tip clearance (*utc*/*mtc*). If X and t_{tc} are not mentioned, it represents the case of channels with zero tip clearance.

pressure, and temperature distributions in the fluid domain. Eq. (4) represents the energy equation for the solid domain and is solved in conjunction with Eqs. (1–3) to obtain the temperature distribution in the solid domain.

$$\nabla \cdot (\rho_f \vec{v}) = 0 \quad (1)$$

$$\vec{v} \cdot \nabla (\rho_f \vec{v}) = -\nabla p + \nabla \cdot (\mu_f \nabla \vec{v}) \quad (2)$$

$$\vec{v} \cdot \nabla (\rho_f C_{pf} T_f) = \nabla \cdot (k_f \nabla T_f) + \phi \quad (3)$$

$$k_s \nabla^2 T_s = 0 \quad (4)$$

The required boundary conditions taken for this study are as follows: a constant velocity and temperature is given as the inlet boundary condition at the top surface of the inlet port, while a pressure outlet is selected as the outlet boundary condition. As mentioned, periodic boundary conditions are applied at the left and right boundaries. The bottom surface of the base substrate, referred to as the heat flux surface in subsequent discussion, is subjected to uniform heat flux boundary condition, while all the remaining (external) walls are chosen as adiabatic.

The SIMPLEC algorithm is employed for solving the governing equations. For better accuracy, second order discretization schemes are chosen for pressure, momentum, and energy. For each case, the simulation is deemed converged when the residuals fell below 1×10^{-6} , or at least the last 20 iterations fell below 5×10^{-4} .

Simulations were performed for straight and oblique microchannels both with and without varying tip clearance. Table 2 shows the numerical values of the geometric variables associated with the oblique microchannels and tip clearances. For direct comparison purposes, channel width and depth, fin width, and number of channels were kept same for straight and oblique microchannel configurations, as shown in Table 1. Two different oblique fin angles (i.e., 40° and 60°) were considered, keeping the oblique flow channel width and fin pitch constant. Two variants of tip clearance were considered: uniform tip clearance (*utc*) and modulated tip clearance (*mtc*), as shown in Fig. 3. It is worth noting that *mtc* was conceived after examining the impact of *utc* on fluid flow occurring through the oblique channels and the associated

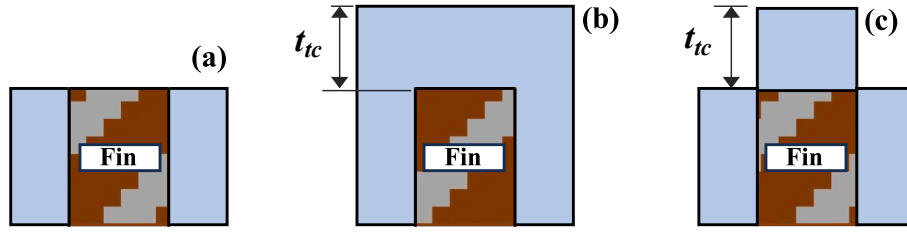


Fig. 3. Schematics showing variants of tip clearance, a) no (zero) tip clearance, b) uniform tip clearance (*utc*) and c) modulated tip clearance (*mtc*).

Table 3

Mesh independency results.

Channel	Cell size in channel's fluid domain (μm)	Total number of cells (million)	Average temperature of the heat flux surface (K)	Relative change in temperature (K)	Inlet pressure (Pa)	Relative change in pressure (%)
<i>Str_utc_100</i>	12.5	7.31	321.87	—	8302	—
	10.0	11.81	322.32	0.45	8335	0.3
	7.8	19.99	322.39	0.07	8343	0.1
<i>Obl_40°_utc_100</i>	12.5	10.59	316.52	—	9453	—
	10.0	11.72	317.05	0.53	9533	0.8
	7.8	20.34	317	0.05	9551	0.2
<i>Obl_40°_tc_100</i>	12.5	11.27	315.05	—	14,746	—
	10.0	12.57	315.60	0.55	14,910	1
	7.8	15.22	315.50	0.1	14,925	0

reasoning is explained later in this paper. The tip clearance was varied until degradation in the thermal performance was observed for both the *utc* and *mtc* cases. Based on these geometric parameters, microchannels were given the listed nomenclature for ease of referencing, as shown in Table 2. For each channel configuration, simulations were performed at three different flow rates i.e., 250, 450, and 650 mL/min while maintaining a constant inlet fluid temperature of $T_{in} = 22^\circ\text{C}$ and a uniform heat flux of $q'' = 75 \text{ W/cm}^2$ at the heat flux surface. Water was chosen as the working fluid with temperature dependent thermo-physical properties and the properties were taken from Ref. [31]. Aluminium with default properties in FLUENT was selected as the solid domain material.

To obtain mesh independence with manageable computational time, a study was carried out by gradually reducing the cell size in the computational domain for both straight and oblique microchannel cases. Particularly, the adequate cell size in the channel fluid domain is critical to accurately capture the heat transfer and hydrodynamic characteristics. Table 3 shows few examples of mesh independence study for the critical case of highest flow rate of 650 mL/min, and with straight and oblique channel configurations. Based on the relative change in the average base temperature and the inlet pressure with mesh size, it is seen that mesh independent results can be obtained with a cell size of $10 \mu\text{m}$ in the channel fluid domain, maintaining a maximum cell size of $100 \mu\text{m}$ in the solid and the inlet/outlet port fluid domains.

2.2. Data reduction

The laminar flow behaviour assumed in the simulation was confirmed by calculating $\text{Re} = \frac{\rho_f v D_h}{\mu_f}$, accounting the bulk fluid temperature. The Re values based on bulk properties were found to be 290, 500, and 720 for the flow rates 250, 450, and 650 mL/min respectively for the channel configurations without tip clearance. The simulation results provide velocity, pressure, and temperature distributions within the computational domain. Based on this data, one can calculate the pressure drop across the microchannels (ΔP) and the average temperature of the heat flux surface ($T_{b,avg-sim}$). ΔP was evaluated by subtracting spatial-averaged pressure values computed at the inlet and outlet boundary faces. $T_{b,avg-sim}$ was calculated by averaging the cell temperature values

on the heat flux surface as shown in Eq. (5), wherein $T_{b,i}$ represents the temperature of i^{th} cell and n indicates number of cells on the heat flux surface. These two parameters were used to calculate the overall thermal resistance, R_{th} , (Eq. (6) and pumping power, P_{pump} , (Eq. (7) which assess the thermal and hydraulic performance of each microchannel configuration with and without tip clearance. Furthermore, a relative thermo-hydraulic performance of every design studied w.r.t. *Str* design was assessed based on a 'Performance Index' (*PI*) parameter, as defined in Eq. (8). The *PI* parameter was defined in the similar way as discussed in Ref. [32], wherein various definitions of the *PI* parameters were compared for predicting the overall performance accurately.

$$T_{b,avg-sim} = \frac{\sum_{i=1}^n T_{b,i}}{n} \quad (5)$$

$$R_{th} = \frac{T_{b,avg-sim} - T_{in}}{q} \quad (6)$$

$$P_{pump} = \Delta P \times V' \quad (7)$$

$$PI = \left(\frac{R_{th}}{R_{th,Str}} \right)^{-1} / \left(\frac{\Delta P}{\Delta P_{Str}} \right)^{1/3} \quad (8)$$

3. Experimental Apparatus

To verify the simulation results, experiments were carried out on in-house test facility. It primarily consisted of a heat source and heat sink test section assembly, flow loop system, and data acquisition system, as depicted in Fig. 4(a) and 4(b). Here, the heat source comprised of standard copper heated meter bar with a ceramic heater (500 W) attached at its bottom to provide the necessary heat load. To avoid heat losses from the copper bar, ensuring one dimensional heat conduction, the meter bar is well isolated from the surrounding with a thick layer of thermal insulation. The straight microchannel heat sink of dimensions $L=50 \text{ mm}$, $W=40 \text{ mm}$ and $t_e = 4 \text{ mm}$ (Fig. 4(c)) was fabricated in-house using a micro-milling machine method and fixed on to the top of the meter bar with a thin layer of thermal paste sandwiched between them. A $20 \text{ mm} \times 20 \text{ mm}$ cut having a depth of 2 mm was machined at the

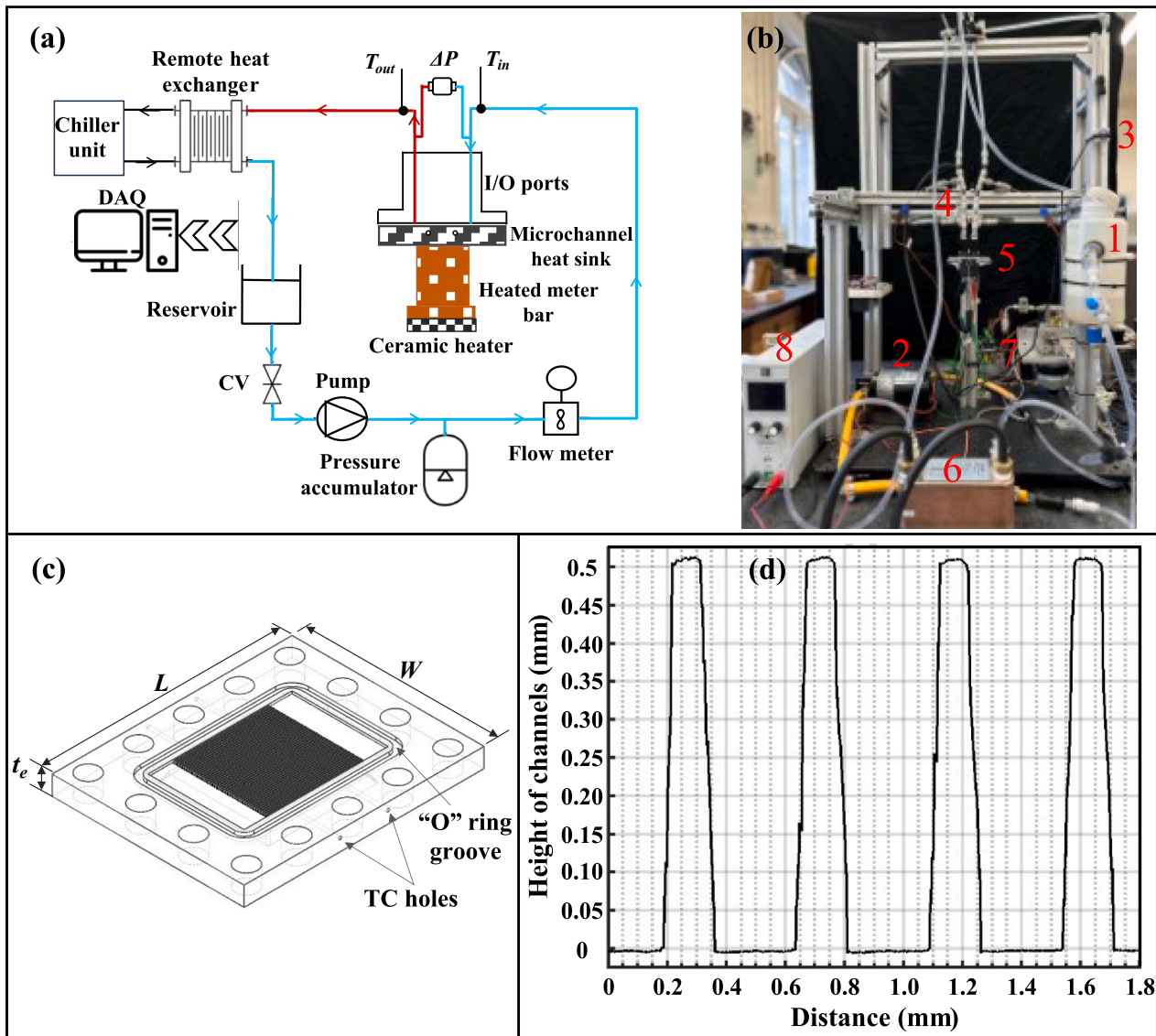


Fig. 4. A) schematic of the experimental flow loop, b) photographic view of the flow loop, c) solid model of the employed heat sink, and d) the shape profiles of the micro-milled channels examined using confocal microscopy. 1-reservoir, 2-pump, 3-flow meter, 4-differential pressure transducer, 5-heat source and heat sink assembly, 6-remote heat exchanger, 7-DAQ, and 8-DC power source.

central portion of the heat sink base in order to clamp the heated meter bar. The dimensions of the milled-microchannels were examined using confocal microscopy. Fig. 4(d) shows the geometry and dimensions of the machined channels, displaying some tapering of the channels and an overall trapezoidal-like shape, as opposed to the ideal rectangular geometry. To measure the base temperature, four *K*-type sheathed thermocouples having 0.5 mm diameter were inserted into the holes drilled in the microchannel plate, the location of which can be seen in Fig. 4(c). The first hole from each side was drilled at 5 mm distance from the start of the channel and the two holes from each side were 10 mm distance apart from each other. The distance between the channel base and the centreline of any hole was kept 1.0 mm. Two of these thermocouples from one side measure the centreline temperature of the plate while other two measure the temperature underneath the extreme side channel.

The flow loop was constructed by connecting a reservoir, pump, pressure accumulator, flow meter, and remote heat exchanger in series with 8 mm Teflon tubing. The microchannel heat sink inlet/outlet (I/O) ports were attached between the flow meter and the remote heat exchanger. A high-pressure diaphragm pump (Flojet: D3135H7011AR)

was used to circulate the flow and a pressure accumulator (Imera, 11 bar) was placed after the pump to minimize any flow fluctuations. A TITAN NSF 845-VOP-O flowmeter calibrated at 22.3 °C with a range from 0.2 to 4.5 LPM was used for measuring the flow rates, with proper straight flow tubing upstream of the flow meter. Two *T*-type thermocouples were inserted in the flow before and after the test section to measure the inlet and outlet coolant temperatures. Adjacent to these thermocouples, pressure ports were installed to which a differential pressure transducer (OMEGA, ± 0.35 bar) was connected to measure the pressure drop across the test section. A flat-plate heat exchanger plumbed to a temperature-controlled chiller was placed in between the microchannel heat sink and reservoir to control and maintain the constant inlet fluid temperature (22 °C) throughout the experiments. National Instruments DAQ hardware (NI cDAQ-9174, NI 9211, and NI 9205) was employed to record time-resolved data from thermocouples, pressure sensor, and flow sensor with the help of in-house LabVIEW program.

Experiments were conducted under the same conditions as those chosen in the simulation for the straight microchannel heat sink without tip clearance. Before initiating the experiments, a high flow rate was

Table 4

Uncertainties associated with the key parameters.

Parameter	Value	Uncertainty
Temperature (T)	—	0.3 K
Differential pressure (ΔP)	0.35 bar	0.05 – 0.4 % FS
Flow rate (V')	0.2 – 4.5 LPM	1.5 % FS
Voltage (V)	52 V	0.1 V
Current (I)	6 A	0.01 A
Supplied power (q'_{total})	315 W	0.3 %
Heat flux surface length/width	20 mm	1 mm
Heat flux surface area (A_s)	20 × 20 mm ²	7 %
Supplied heat flux ($\frac{q'_{total}}{A_s}$)	75 W/cm ²	7 %
R_{th}	0.098 – 0.138 (K/W)	1.13 % – 0.83 %
P_{pump}	0.011 – 0.106 (W)	27 % – 10.5 %

maintained in the flow loop to expel trapped air in the tubes. Subsequently, a flow rate of 250 mL/min was maintained and the LabVIEW program was run to monitor and record the temperatures, pressure drop and flow rate. Then, the ceramic heater was energized to generate the uniform heat at a constant power of 315 W, which is slightly higher than that considered in the simulation. This was intentionally done to account for the heat losses from the heat source assembly. Once inlet and outlet liquid temperatures along with heat sink temperature showed steady behaviour with time, the temperature, pressure, and flow rate data were captured for 5 min with a sampling rate of 5 samples/sec. Following this, the flow rate was increased, and the same procedure was repeated. It is to be noted that the time-average of the recorded data for each parameter was used for further analysis and verification purposes.

During experiments, although the heat source was thermally insulated from the surrounding, it loses a part of the supplied heat to the surrounding. Therefore, it becomes essential to calculate the net heat supplied to the test section (q'_{net}). In the present work, it was calculated as given in Eq. (9). On an average, the heat loss (q'_{loss}) was found to be 10 % of the total heat generated by the heater (q'_{total}) over the range of flow rates. Considering this, the net heat flux subjected to the heat sink was found to be approx. 71 W/cm² as opposed to 75 W/cm² considered in the simulation. The spatial averaged base temperature ($T_{b,avg-exp}$) of the microchannel plate was calculated by Eq. (10), wherein $T_{b-exp,i}$ ($i = [1, 4]$) denotes the temperature measured by the i^{th} thermocouple fixed in the microchannel base plate, as discussed earlier. The R_{th} and P_{pump} were calculated using Eqs. (6) and (7) respectively but employing experimentally measured parameters. The pressure drop (ΔP) as measured by the pressure transducer was employed in Eq. (7) because the pressure taps were very close to the inlet/outlet ports.

$$q'_{loss} = q'_{total} - q'_{net} = \{V \times I\} - \{m' C_p (T_{out} - T_{in})\}; \quad (9)$$

$$T_{b,avg-exp} = \frac{\sum_{i=1}^4 T_{b-exp,i}}{4} \quad (10)$$

Uncertainties involved in the measurement of temperature, pressure, flow rate depend on the corresponding sensor quality and are given in Table 4. The uncertainties associated with the evaluated parameters including supplied power, heat flux area, supplied heat flux, R_{th} and P_{pump} were estimated using a standard error propagation formula (Eq. (11)) and consequently presented in Table 4.

$$\delta y = \sqrt{\sum_{i=1}^n \left(\frac{\partial y}{\partial x_i} \delta x_i \right)^2} \quad (11)$$

In Eq. (11), $y = f(x_i)$ is the dependent variable and x_i represents i^{th} independent variable. In the present work, y represents the evaluated parameters like supplied heat power, heat flux surface area, supplied heat flux, R_{th} , and P_{pump} . The uncertainty calculation of P_{pump} is shown in Eq. (12) for the reference, considering Eq. (7).

$$\delta P_{pump} = \sqrt{\left(\frac{\partial P_{pump}}{\partial \Delta P} \delta(\Delta P) \right)^2 + \left(\frac{\partial P_{pump}}{\partial V'} \delta(V') \right)^2} \quad (12)$$

To verify the accuracy of the simulation, a separate 3D geometry of the microchannels was modelled to approximately replicate the geometry of the milled microchannels. For verification purpose, experimentally evaluated overall thermal resistance and pumping power values were compared with the simulation predictions and subsequently presented in Fig. 5. It is clear from the figure that both the overall thermal resistance and pumping power values estimated numerically follow similar trends as a function of flow rate with that obtained with the experimental results. There is a maximum discrepancy of 4 % in the thermal resistance and 18 % in the pumping power between simulation and experimental data. The observed discrepancy in the thermal resistance is adequately close to the experimental uncertainty, considering some assumptions had to be made with regard to the channel geometry. As far as discrepancy in the pumping power is concerned, it is also well within the margins of uncertainty associated with the pumping power, which varies from 11–27 % as shown in Table 4. Nonetheless, any small differences in the channel geometry considered for simulation from its experimental counterpart tend to significantly influence the pumping power [33]. It is to be noted that the discrepancy in both parameters is seen to be increasing with flow rate. This is mainly because the influencing factors including any small deviation in the channel geometry considered for simulation from its experimental counterpart, neglected effects of surface roughness and machining artefacts in the simulation

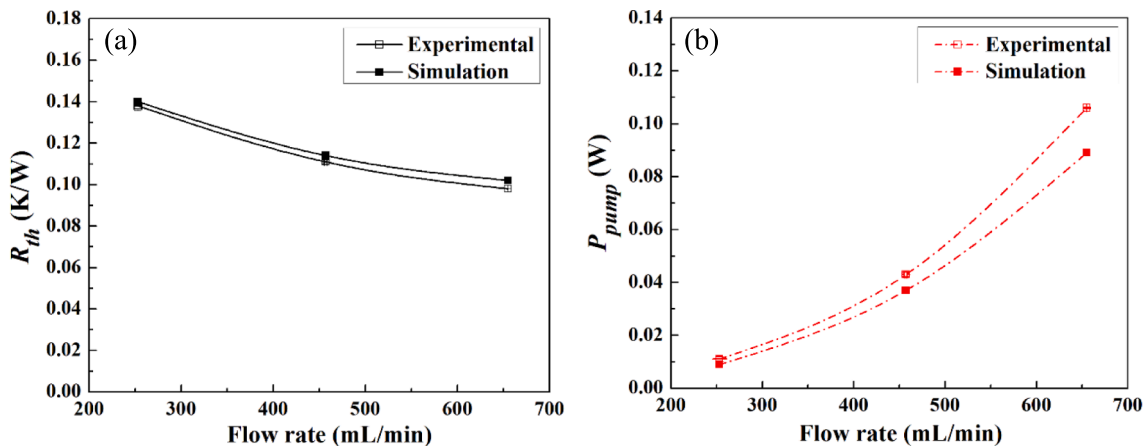


Fig. 5. Comparison of numerical and experimental data; a) Overall thermal resistance, R_{th} , and b) Pumping power, P_{pump} , for varying flow rates. The error bars indicate repeatability of experiments showing 95% confidence interval data.

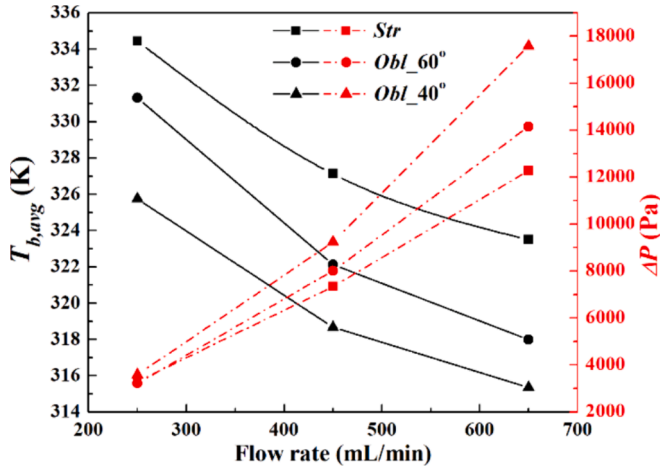


Fig. 6. Variations of $T_{b,avg}$ and ΔP as a function of flow rate for straight and oblique channel configurations.

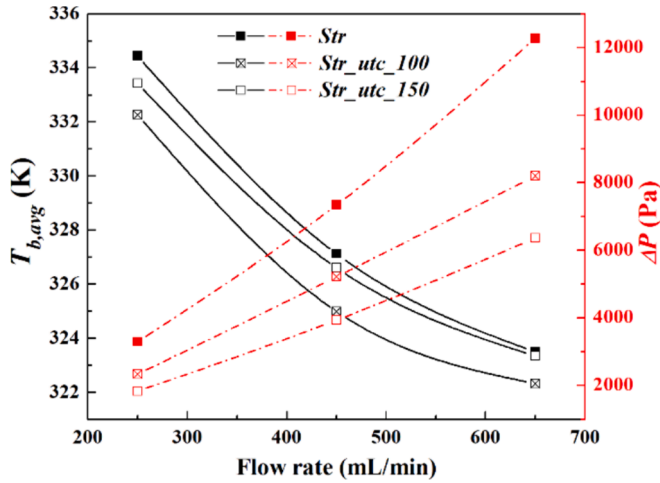


Fig. 7. Variations of $T_{b,avg}$ and ΔP as a function of flow rate for straight microchannels with and without *utc*.

become more sensitive with increasing flow rate [33].

4. Results and discussion

4.1. Uniform tip clearance

Fig. 6 illustrates the variations in the average base temperature and pressure drop for increasing flow rate for the straight microchannel, as

well as those with 40° and 60° oblique fin angles, without tip clearance. The observed trends indicate a decrease in base temperature and an increase in pressure drop with increasing flow rates for all channels, as anticipated. Notably, the *Obl_40°* configuration exhibits the lowest base temperature, while the straight channels consistently yield the highest base temperature across all flow rates. This observation is attributed to the enhanced fluid mixing and periodic breakage and redevelopment of thermal boundary layer caused by the oblique slots, which improves heat transfer and lowers the base temperature in the case of Oblique channels. Between *Obl_40°* and *Obl_60°*, it is found that these two phenomena are more intense in *Obl_40°*. It is to be noted that these findings related to zero *tc* oblique microchannels are adequately reported in the open literature [20,23] and thus discussed here only in brief. The reduction in temperature below the straight channel baseline is observed to be in the region of 8 K for *Obl_40°*, representing a reduction in the thermal resistance of about 28 %. Thus, both oblique configurations are found to decrease the thermal resistance compared with the straight microchannel, though at the expense of an increased pressure drop. This increases with lower oblique cut angle and increased flow rate, up to 40 % for *Obl_40°* over *Str* at the highest flow rate (see Fig. 6). These observations qualitatively align with previous findings reported in the open literature [22,23].

The impact of uniform tip clearance (*utc*) on the aforementioned test cases is examined for both straight and oblique channels, with tip clearance heights of 100 μm and 150 μm . The corresponding data is presented in Figs. 7 and 8, respectively, for straight and oblique channels. As illustrated in Fig. 7, the tested *utcs* are observed to reduce the base temperature in conjunction with a significant decrease in pressure drop. Specifically, the 100 μm *utc* yields the highest base temperature reduction (1 K - 2 K), while it is less pronounced for the 150 μm *utc* case. Conversely, the pressure drop data displays a monotonically decreasing trend with increasing values of *utc*, as expected. Notably, up to 50 % reduction in pressure drop is observed for 150 μm tip clearance case, and this is achieved with minimal influence on the thermal resistance.

The reduction in base temperature and consequent enhancement in thermal performance with *utc* for straight channels can be understood as follows: heat transfer in microchannels is influenced by the convective surface area and intensity of convective heat transfer. The *utc* enables the fluid to flow over the tip of the fins (channel walls), thereby facilitating additional surface area, since the upper casing material in this scenario is insulative and the fins are of fixed height. However, it also increases the effective cross sectional flow area of the channels, leading to a reduction in channel flow velocity and convective heat transfer coefficient. Consequently, the thermal performance of straight channels with *utc* is dictated by a trade-off between additional convective heat transfer in the tip clearance area and reduced convective heat transfer intensity within the channels. For *utc* values of 100 μm and 150 μm , the enhanced convective heat transfer in the tip clearance area tends to outweigh the reduction in convective heat transfer intensity in the channels, resulting in an overall reduction in the thermal resistance.

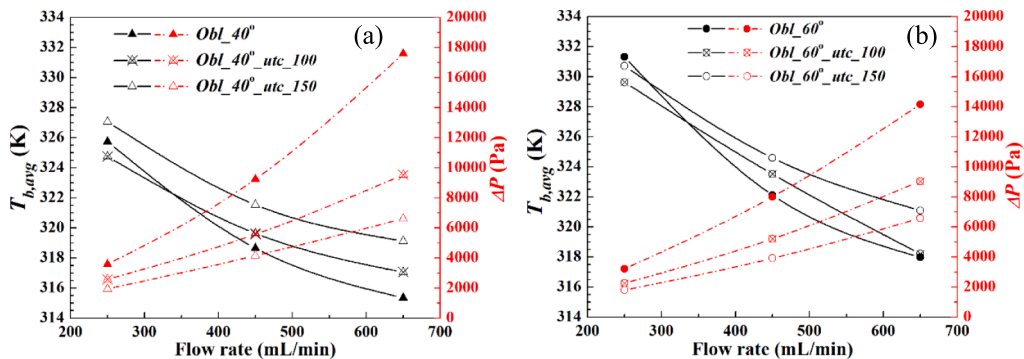


Fig. 8. Variations of $T_{b,avg}$ and ΔP as a function of flow rate for a) *Obl_40°* and b) *Obl_60°* channels with and without *utc*.

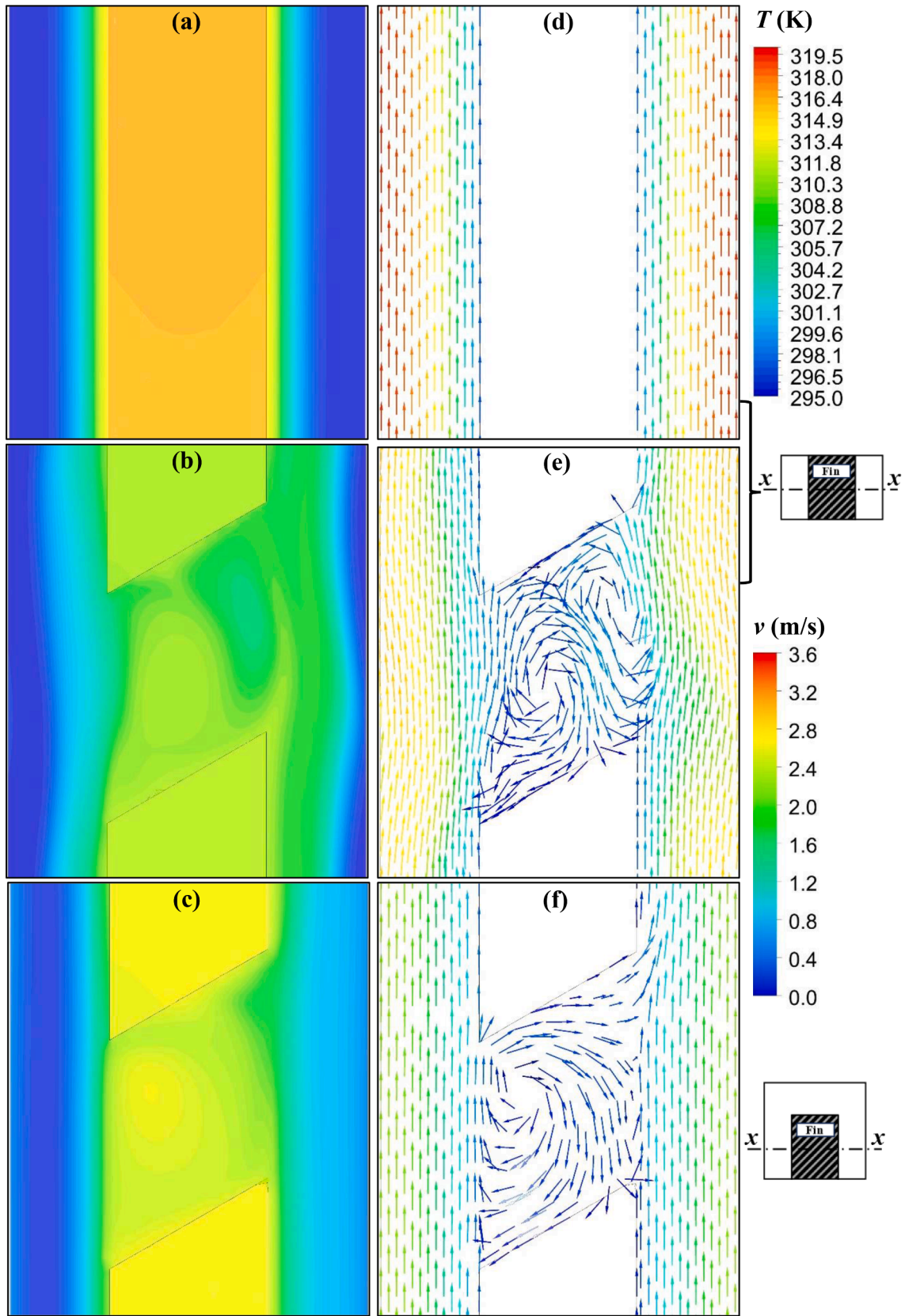


Fig. 9. Temperature contour and velocity vector maps respectively for the case of a) and d) *Str*, b) and e) *Obl 60°*, and c) and f) *Obl 60°_utcl50* at 650 mL/min. These maps are generated at the midplane of fin height, indicated by x-x in the associated schematics, and the data is shown for the 10th oblique slot from the entrance.

Thus, taken together with the significant reduction in the pressure drop and the fact that it does not represent a notable manufacturing challenge, the findings presented in Fig. 7 corroborates the effectiveness of *utc* for straight channels.

To assess the efficacy of *utc* for oblique channels, the corresponding base temperature and pressure drop data are presented in Fig. 8. For *Obl* 40° cases (Fig. 8(a)), the base temperature is found to be higher with both tip clearance levels except at the lowest flow rate, where the 100 μm *utc* show minor (~ 1 K) temperature reduction relative to the no tip clearance case. The increase in base temperature with tip clearance is more pronounced for the 150 μm *utc*. Fig. 8(b) shows similar observations regarding base temperature for *Obl* 60° , with a minor improvement at the lowest flow rate (~ 2 K), though consistently poorer heat transfer performance over the rest of the test scenarios relative to the no tip clearance case. Both datasets in Fig. 8 indicate that *utc* generally increases the thermal resistance for oblique fin channels, particularly at flow rates above 250 mL/min. However, it is worth noting that the generally increased thermal resistance with tip clearance is moderate, reaching of 15 % for *Obl* 40° _*utc*_150 and 10 % for *Obl* 60° _*utc*_150 at the highest flow rate. Therefore, unlike the straight channel, the trade-off between increased surface area of the channel tips and the reduced convective heat transfer intensity in the lower channels for oblique channels tends to be in favour of the reduced convective intensity, thus diminishing the overall cooling effectiveness, albeit marginally. Despite this, Fig. 8 shows that the pressure drop is substantially lowered with increasing *utc* for both oblique fin angle channels, especially at 650 mL/min, ranging from 45–65 % and 36–53 % for *Obl* 40° and *Obl* 60° respectively.

The discussion above has revealed that the inclusion of oblique cuts in the channel walls can notably enhance the overall heat transfer compared to the straight channels, though unlike straight channels, the addition of tip clearance generally deteriorates the heat transfer. In oblique fin channels, the convective heat transfer is additionally influenced by secondary flows occurring through the oblique slots, which enhances fluid mixing and consequently the heat transfer. For illustration purposes, Fig. 9 depicts the temperature contour plots and velocity vector plots for *Str*, *Obl* 60° and *Obl* 60° _*utc*_150 cases. For a direct comparison, the 10th oblique slot from the inlet is selected for both oblique cases. It is evident that the oblique slot induces vigorous fluid mixing with two oppositely rotating vortices for the case of *Obl* 60° (Fig. 9(b) and 9(e)). In addition, these vortices interact with the main channel flow, disturbing the thermal and flow fields in the channel regions adjacent to oblique slots when compared to those observed with *Str* case (Fig. 9(a) and 9(d)). Moreover, the temperature field shows that fluid mixing in oblique slots reduce the upstream liquid temperature adjacent to the fin walls and thus facilitates relatively colder liquid to the downstream fins, enabling a re-development of the thermal boundary layer. Considering that the overall wetted surface area is slightly smaller (6 %) for *Obl* 60° compared with *Str*, the coupled phenomena intensify the convective heat transfer to the extent that it more than overcomes this surface area reduction, resulting in a substantial heat transfer enhancement over the straight microchannel case. However, for the *Obl* 60° _*utc*_150 case, because the flow bypasses through tip clearance region, less intense fluid mixing is observed in the cross-cut sections, with only one distinct but weak vortex that does not penetrate into the main channel flow, which itself is of comparatively low velocity and higher temperature, as seen in Fig. 9(c) and 9(f). Despite the higher heat transfer surface area of *Obl* 60° _*utc*_150 compared with *Obl* 60° , the disproportionate decrease in the convective heat transfer intensity results in an overall increase in the thermal resistance when tip clearance is added with the oblique microchannels.

4.2. Modulated tip clearance

As discussed earlier, uniform tip clearance (*utc*) weakens the intensity of convective heat transfer in oblique slots owing to the increased

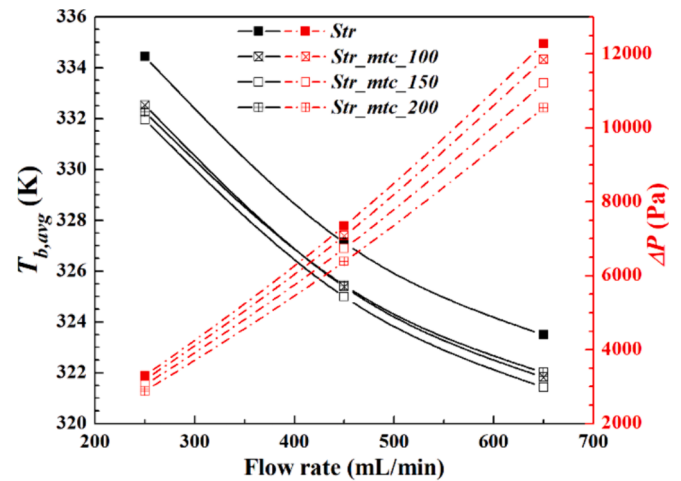


Fig. 10. Variations of $T_{b,avg}$ and ΔP as a function of flow rate for straight microchannels with and without *mtc*.

flow area reducing the channel flow. The convective intensity is reduced to the extent that the overall thermal resistance increases despite a non-insignificant increase in the heat transfer surface area. To counteract these effects, a modified version of tip clearance, namely modulated tip clearance (*mtc*), is introduced, as depicted in Fig. 3(c). The aim of this design is simply to still allow flow over the tips of the channel whilst moderating the associated drop in flow velocity by reducing the impact on the flow cross-sectional area. It was anticipated that this could retain the positive aspects of higher surface area and lower pressure drop, whilst sustaining high intensity of convective heat transfer.

The *mtc* influence on base temperature and pressure drop is compared with those of the no tip clearance cases for both straight and oblique channels in Figs. 10 and 11, respectively. It is observed in Fig. 10 that applying *mtc* in straight channels improves the cooling performance. Referring back to Fig. 7, the sensitivity to tip clearance height is now quite low compared to that of the *utc* design, with the surface temperatures for the *mtc* cases approximately that of *Str*_*utc*_100. The lack of sensitivity to clearance height is indicative of the additional surface area, which is constant between the three *mtc* cases, being the main contributor to the improved heat transfer. It is noted that pressure drop decreases with increasing value of *mtc*, as expected. However, the reduction in pressure drop relative to the no tip clearance is modest (e. g., 8 % for *Str*_*mtc*_150) when compared with the *utc* cases (refer to Fig. 7).

Noting that a key aim of the *mtc* design concept was to improve the convective heat transfer intensity compared to the noticeably deteriorated with *utc* design for oblique fins, Fig. 11(a) and 11(b) show a that indeed this is the case. Comparing with Fig. 8, the figures now indicate a general improvement in cooling performance, with some minor exceptions, whereas the opposite was true for the *utc* test cases. This observation can be explained by considering the temperature and velocity vector plots associated with oblique slots for *Obl* 60° and *Obl* 60° _*mtc*_150, as shown in Fig. 12. The plots clearly illustrate that the general flow structures along with velocity magnitudes in the main channel and within the oblique cut section are comparable for both cases.

Like the straight microchannel case, Fig. 11(a) and 11(b) also show that the oblique channels have a comparatively low sensitivity of the cooling performance to the clearance height for the *mtc* cases, in particular compared to the *utc* cases (see Fig. 8). The *mtc* thermal resistances are typically within 10 % of the no tip clearance case. Generally speaking, since the *mtc* design results in comparable flow patterns (Fig. 12), the lower sensitivity to tip clearance makes sense in the context of the increased surface area being the dominant contributor to the enhanced heat transfer. Similarly, like the straight channels case

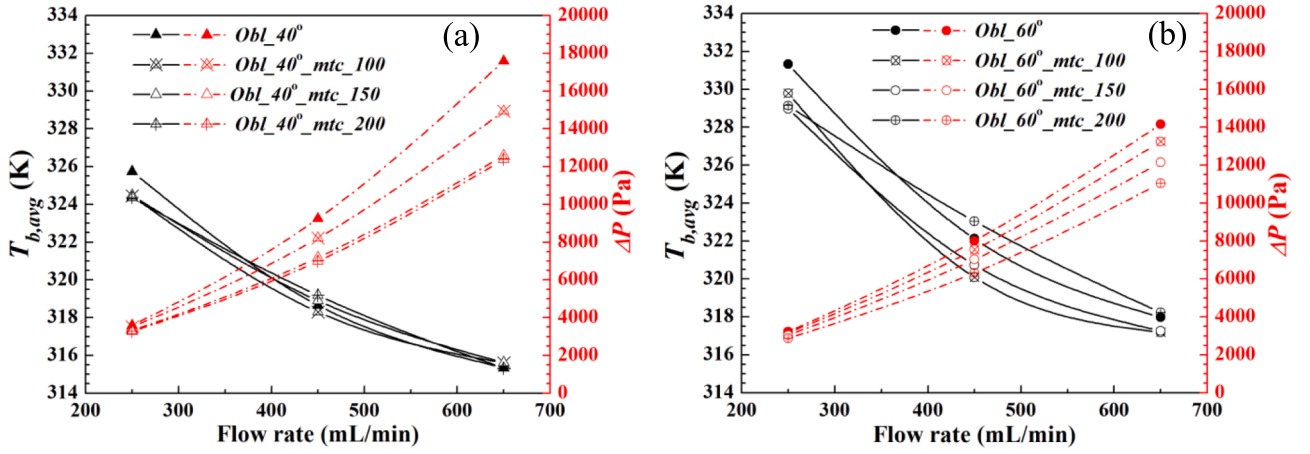


Fig. 11. Variations of $T_{b,avg}$ and ΔP as a function of flow rate for a) Obl_{40° and b) Obl_{60° channels with and without *mtc*.

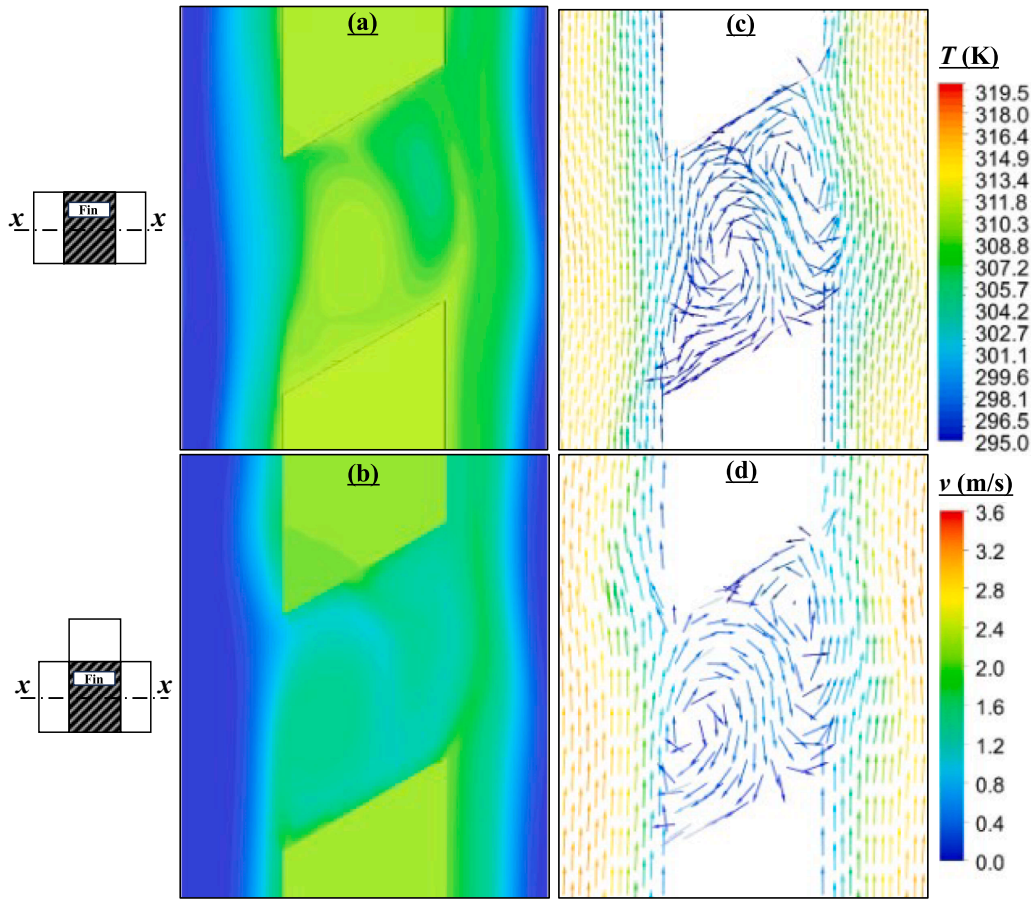


Fig. 12. Temperature contour and velocity vector maps respectively for the case of a) and c) Obl_{60° , and b) and d) $Obl_{60^\circ_mtc_150}$ at 650 mL/min. These maps are generated at the midplane of fin height, indicated by x-x in the associated schematics, and the data is shown for the 10th oblique slot from the entrance.

discussed with *mtc*, the pressure drop reductions are relatively small for oblique channels with *mtc* compared to what was achieved for the *utc* design.

In design situations where one must carefully consider power consumption of the cooling system, it is worth assessing the overall design potential in terms of both the thermal resistance and pumping power penalty. Fig. 13 (a) shows that in general terms, for a given pumping power straight microchannels produce the worst heat transfer, Obl_{60° channels offer improved heat transfer, and Obl_{40° channels render highest heat transfer, and this is true for both with and without tip

clearance configurations. Evidently, the simple addition of tip clearance, regardless of the type, is seen to notably improve the overall thermal-hydraulic performance. In terms of the comparing and contrasting *utc* versus *mtc* designs, for the range of parameters tested here they are similar, with *utc* being marginally better at lower flow rates, and *mtc* outperforming at higher flow rates, and both worthy candidates for enhanced microchannel heat exchangers.

Fig. 13(b) demonstrates the variation of *PI* parameter, indicating relative performance of all the tested channels compared to the straight one without *tc*. As can be seen in the figure, oblique channels without *tc*

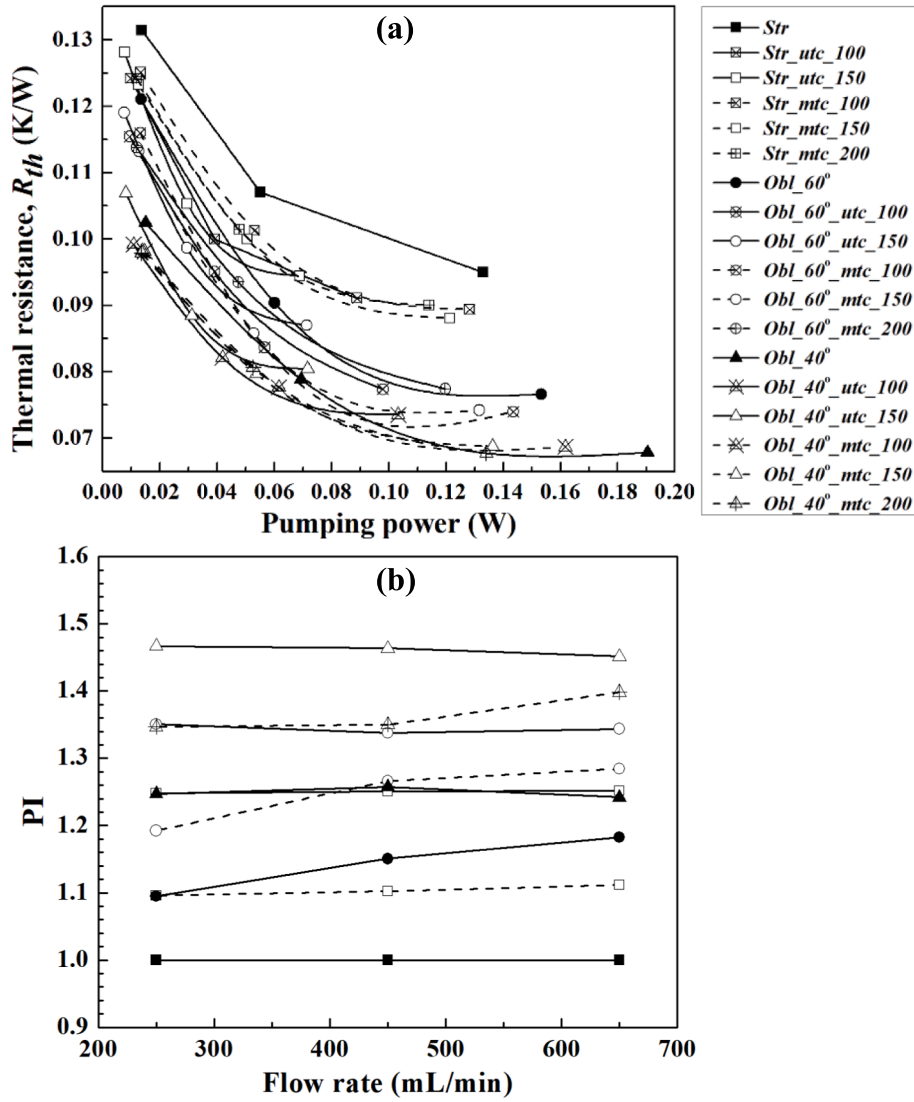


Fig. 13. (a) Variation in overall thermal resistance, R_{th} , as a function of pumping power, P_{pump} , for all the design configurations studied and (b) PI for the selected design configurations which yielded maximum values.

Table 5

Threshold of tip clearance height relative to critical geometric parameters that results in maximum PI .

Channel type	t_c/w_c		t_c/w_f		t_c/t_c	
	UTC	mtc	UTC	mtc	UTC	mtc
Straight	0.6	0.6	0.75	0.75	0.3	0.3
Oblique_60°	0.6	0.6	0.75	0.75	0.3	0.3
Oblique_40°	0.6	0.8	0.75	1.0	0.3	0.4

improve PI by almost 10–18 % and approx. 25 % with 60° and 40° angles respectively across the flow rates when compared to that of *Str*. However, it is interesting to note that mere introducing 150 μm *UTC* for *Str* results in the same PI as observed with *Obl_40°*. It is clearly visible that *UTC* outperforms *mtc* for both straight and oblique channels by 10–15 % as far as PI is concerned though this margin reduces drastically at high flow rates.

It is important to note that the impact of tip clearance height (t_c) on overall performance depends on critical geometric parameter of microchannel such as channel width (w_c), fin width (w_f), and fin height (t_c). To understand this dependence, three different ratios based on the tip clearance height and these geometric parameters are evaluated and

presented their values in Table 5. These values are presented for the cases where the maximum Performance Index (PI) was achieved for each type of the channel with different tip clearance designs. Thus, Table 5 serves as a reference for designing tip clearance strategies for both straight and oblique channels, considering the geometric parameters of the channels to fully realize their performance. It should be noted, however, that the maximum PI was achieved at the largest tip clearance height considered in this study for *UTC* across all channels, warranting further investigation. Additionally, it is essential to mention that these reference values are valid specifically to aluminium-based heat sinks using water as the working fluid. The data presented in Fig. 13 and Table 5 thus motivate the next step in this research programme, which will involve a systematic multi-objective optimization study of both tip clearance design concepts over a broader and more highly resolved range of the geometric design parameters.

5. Conclusions

Considering the limited explorations of tip clearance for enhanced heat sinks, the influence of tip clearance on thermal and hydraulic performance of both straight and oblique microchannels was investigated through numerical simulations. Water with temperature

dependent properties was selected as the working fluid along with aluminium as the heat sink material. Two variants of tip clearance, namely conventional/uniform tip clearance (*utc*) and modulated tip clearance (*mtc*), were explored for straight microchannels as well as those with 40° and 60° oblique cut fins. The simulations were carried out for three flow rates (250, 450, and 650 mL/min) and different heights of the tip clearance for both types of channels. The velocity and temperature distribution contour plots along the channel were presented and discussed.

In the absence of tip clearance, oblique microchannels were found to enhance thermal performance at the expense of increased pressure drop compared to straight channels, a result consistent with the existing literature. However, when *utc* was introduced, straight microchannels exhibited improved thermal performance alongside a reduced pressure drop. Specifically, both 100 µm and 150 µm *utcs* showed lower thermal resistance compared to straight channels without tip clearance. Conversely, for oblique channels with *utc*, there was either no change or a significant deterioration in thermal performance, despite a notable reduction in the pressure drop relative to configurations without tip clearance. This decline in thermal performance was attributed to the disproportionate decrease in the convective heat transfer intensity compared to increased heat transfer area due to the provision of uniform tip clearance.

With the anticipation of partially mitigating these effects, modulated tip clearance was conceptualized and introduced for the oblique channels. In general, all channels, including straight and oblique configurations with *mtc* exhibited improved thermal performance, depending on the magnitude of *mtc* and the oblique fin angle, with relatively less reduction in pressure drop when compared to that with *utc* cases. Considering both heat transfer augmentation and associated hydraulic power penalties, *utc* performed better at lower flow rates, while *mtc* outperformed *utc* for higher flow. However, as far as *PI* was concerned, *utc* design led to better improvement in the overall performance compared to that with *mtc* for both channels, depending on the flow rates. Furthermore, maximum permissible relative values of the tip clearance height compared to the critical geometric parameters were given, which act as guidelines for designing tip clearance strategy. This highlights the potential of both *utc* and *mtc* designs in oblique cut microchannels and warrants further investigation that implements multi-objective optimization strategies to mathematically minimize both the thermal resistance and required hydraulic power simultaneously. It should be noted that while the present work systematically examines tip clearance designs for enhanced heat sinks, it has certain limitations. These include neglected effects of flow distribution across the channels, the limited range of geometric parameters considered, and the limited number of experiments conducted. These further warrants more experimental investigations and numerical optimization studies on full-scale heat sinks.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This publication was developed with the financial support of Science Foundation Ireland (SFI) under grants numbered 13/RC/2077 and 12/RC/2278.

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